MCA SEMESTER-I

School of Computer and Systems Sciences,
Jawaharlal Nehru University, New Delhi -110067

Computer Fundamental (CS-402)

Total Marks: 20 **Date: 07.02.2022** **Time: 2hr**

All Compulsory

5. Express the following function in a) some of minterms and b) product of maxterms.
   d) F(x,y,z)=1
   e) F(w,x,y,z)=y'z+wxy'+wxz'+w'x'z
   f) F(x,y,z)=(xy+z)(y+xz) 5

6. For the function ; F=xy+xy'+y'z
   e) Obtain the truth table
   f) Implement the function using 2-input gates
   g) Implement the function using 2-input gates after minimizing.
   h) Compare the number of gate required in b) and c) 5

7. Implement 3-bit Odd function. 5

8. Simplify the following function F using don't care condition *d* in :
   F=w'(x'y+x'y'+xyz)+x'z'(y+w)
   D=w'x(y'z+yz)+wyz
   - Implement the function using 2-input gates 5

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Computer Fundamental (CS-402) Mid-I

Total Marks: 20 Date: 08.09.2024 Time: 2hr

All Compulsory

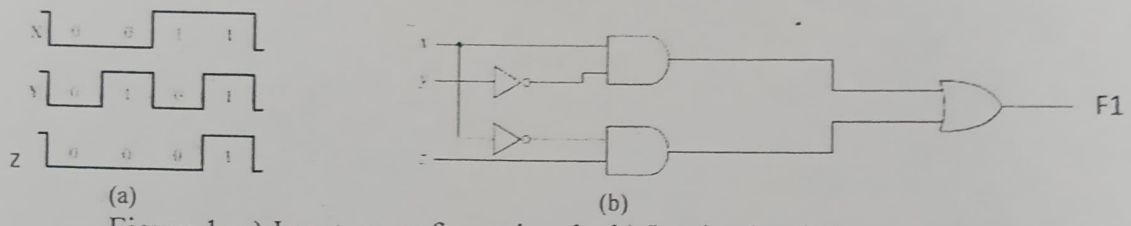
1. Convert the Hexadecimal 2AC5 to decimal, octal and Binary bases. 2
2. Simplify the Boolean function to a minimum number of literals: $x'y' + xy + x'y$ 2
3. Find the complement of : $(x+y'+z)(x'+z')(x+y)$ 2
4. Implement half subtractor. 2
5. Simplify the function: $F=\Sigma(1, 4, 5, 6, 12, 14, 15)$. Implement the function without simplification and after simplifying to check the number of gates in each case. 4
6. Implement BCD to excess three code converter. 4
7. Implement Full adder using decoder. 4

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Total Marks: 20 Date: 07.10.2025 Time: 2hr

All Compulsory

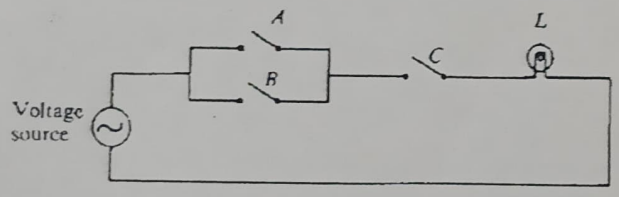
i. a) By means of a timing diagram similar to Fig. 1(a), show the signals of the outputs F1 in Fig.1(b) as functions of the inputs x, y and z.



(a) (b)

Figure 1: a) Input waveform signals b) Logic circuit

b) Express the switching circuit in binary logic notation.



2.5+2.5=5

2. Design a combinational circuit for BCD-Excess-3 coder converter. 5

3. Design BCD-to-decimal decoder. 5

4. Design 4 to 2 Priority encoder. 5

043.

School of Computer and Systems Sciences
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Mid Term Examination-Academic Year 2026-27

Program	:	MCA	Semester	:	I
Course Title	:	Computer Fundamentals	Course Code	:	CS-402
Faculty Name	:	Subhash Chandra Patel	Date/Time	:	01-10-2026/2:00PM-3:00 PM

Attempt any four (04), each question carry equal marks

1. **(A) Simplify the Boolean expression** 2.5

$$F=A(B+C)+A'B$$

using Boolean laws. Explain why each step is valid.

(B) Compare two expressions 2.5

Determine whether the following expressions are equivalent:

$$F=A+B'C \text{ and } G=(A+B')(A+C)$$

Give a mathematical justification.

2. **(A) Prove or disprove** 2.5

Is the following Boolean identity always true?

$$A+BC=(A+B)(A+C)$$

Explain your reasoning.

(B) Implements a digital circuit for given function 2.5

$$F=AB+AC+BC.$$

Explain how you could reduce the number of gates required while preserving the same output.

3. A circuit has the Boolean function 05

$$F=(A+B+C)(A+B'+C)(A'+B+C).$$

Simplify the expression without constructing a complete truth table.

4. Prove using Boolean algebra that $(A+B)(A+C)(B+C)=AB+AC+BC$. 05
5. Minimize the following Boolean function- 05

$$F(A, B, C, D) = \Sigma m(0, 1, 3, 7, 8, 10, 11, 13, 15)+^{5}\Sigma d(2,4,5,9)$$

6. Find a Boolean expression containing at least three variables that is **equivalent to its own complement after one variable is complemented**. Explain your construction.

**********Good Luck**********